

Designing immersive virtual reality stories with rich characters and high interactivity to promote learning of complex immunology concepts

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ABSTRACT

Immunology concepts taught at the college level often involve complex factual and process-based knowledge and can be difficult for students to visualize and relate to. To promote students' understanding of specific immunology concepts and increase their learning motivation and engagement, we leveraged the strengths of immersive storytelling and virtual reality (VR) interactivity and created a guided interactive and immersive VR storytelling experience with rich characters and engaging narratives. The experience builds on immunology concepts of a specific type of white blood cell called neutrophils, and it takes students on an exciting journey inside the human body by embodying the cell and fighting pathogens. In this demo paper, we will describe the design of a working prototype of the VR application.

Keywords: Virtual reality, storytelling, immersion, interactivity, education, immunology.

Index Terms: Human-centered computing – Human computer interaction (HCI) – Interaction paradigms – Virtual reality; Human-centered computing – Human computer interaction (HCI) – Interactive systems and tools

1 INTRODUCTION

Immunology concepts taught at the college level are inherently complex, abstract, and process-based, involving hard factual knowledge and jargon of the field [1] and presenting learning barriers to college students. Immunology knowledge can be difficult for students to visualize and relate to. In our communication with faculty and students in the Biological Sciences Department at Virginia Tech (VT), we have discovered that current methods of teaching immunology, especially in entry-level orientation courses, usually involve didactic lectures in large-size classes and rote learning for in-class drills, which do not promote learning in an active way for most students. These issues in biology education call for new instructional tools and mediums that can effectively engage students into learning.

Practices and studies of using virtual reality (VR) to promote learning have been abundant in the fields of education and instructional design since the 1990s. In a review of VR applications in education, Slater and Sanchez-Vives [2] concluded that VR is an excellent tool for education for several major reasons. For example, VR helps transform the abstract into the tangible, which is extremely useful in learning abstract concepts like mathematics and geometry. It provides hands-on practice opportunities by allowing a learner to learn from doing than just watching. It also makes it

possible to situate learning experiences in places that are impossible or hard to reach and creates novel experiences for the learner.

In classrooms, digital storytelling is frequently used as a pedagogical tool to engage students in learning. Particularly, educators have used stories in science teaching to make subjects accessible to novice learners. Studies have shown that storytelling in science education promotes students' understanding of complex science concepts (e.g., theory of relativity [3, 4]). In addition, story types may also have an impact on learning engagement and outcome. O'Neill [5] developed a story typology that utilizes different story types for instructional purposes in an organizational setting. The typology suggests that epic stories with high color (rich in structural elements and plot content) and high need fulfillment make an effective story structure for instructional purposes, meeting both storyteller and listeners' needs.

Based on these findings, we hypothesize that an immersive digital storytelling experience in VR with a high-level design of story elements like rich characters and a complete story arc, blended with a good amount of user interactivity, may promote immunology learning in several ways:

- Rich interactivity provided by VR promotes active learning by doing and playing.
- Rich characters and a dramatic story arc bring the learners into the story world at a deeper level and engage them more with the information and content carried by the story.
- An embodied learning scenario situates the learners inside the human body at a microscopic scale and creates a novel learning experience.

Driven by the hypothesis, we developed a working prototype of the immunology VR learning application with a high level of storytelling and interactivity. Our overarching goal is to use custom storytelling to explain complex immunology concepts to the learners while at the same time allowing them to reinforce what they have learned with VR interactivity.

2 RELATED WORK

Although storytelling in immersive VR is not a completely new practice, a boom in projects and practices specifically dedicated to the topic has started to take place in the past five years. One noteworthy early example of storytelling in VR was a Disney attraction ride based on the film "Aladdin" [6]. In the experience, the user follows a narrative storyline and flies a magic carpet through a virtual world. This project was an early interactive VR storytelling attempt designed for a completely computer-generated virtual environment. It integrated both storytelling and interactivity into an engaging user experience and is one of the inspirations for our current design. After that, very few applications with similar designs were found in practice. However, in recent years, cinematic VR storytelling has been embraced by filmmakers in the form of either 360-degree videos [7] or computer-generated (CG) animated short films. In such experiences, design focus has been placed on

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strong narratives with rich story elements. One example of such a VR storytelling experience is an award-winning animated music video, *Pearl*, in which the user acts as a ghost viewer who sits in the passenger seat of a 1970 hatchback and watches how a single father tries to raise his young daughter from a child to a teenager, and experiences joys and sorrows in life with her. Although cinematic VR storytelling has the potential for teaching and learning, one major drawback of the medium is that most of them only provide passive viewing experiences and lack the interactivity that supports the process of knowledge construction (e.g., allowing users to change viewpoints or interact with characters in the scene). We hypothesize that passive VR learning experiences might lead to a loss of engagement if a user is exposed to the medium very often or for a long time. Thus, we aim to provide immersive storytelling that is also highly interactive in our current designs.

3 EXPERIENCE DESIGN

3.1 Immunology Concept and Learning Objectives

We started the project by first communicating with a professor and her graduate students in the Biological Sciences Department at Virginia Tech to identify existing learning barriers in classrooms. From our conversations, we identified one concept area in immunology that a lot of undergraduate freshmen had problems with. The concept is about a specific type of white blood cells called neutrophils. Neutrophils are the most abundant white blood cells in the human immune system. Their major functions in the immune defense process involve fast transmigration from blood vessels into an infection site in body tissues at the onset of the infection and using three different mechanisms—degranulation, phagocytosis, and neutrophil extracellular traps (NETS)—to kill bacteria there. This specific learning concept involves factual, conceptual, and process-based knowledge and presents learning difficulties to students in classroom instruction. In addition, our interviews with students suggested that extensive and unfamiliar terminology is another issue preventing them from fully understanding the concept. Finally, limited images in textbooks do little to help them understand and contextualize the ways that neutrophils work. Based on the chosen concept, we then worked with the immunology professor and developed the following learning objectives for the VR experience:

- Students need to understand the steps and process of neutrophil transmigration.
- Students need to understand the function of selectin binding during neutrophil transmigration.
- Students need to understand the differences between three major killing mechanisms (degranulation, phagocytosis, and NETS) and how each of them affects body tissue inflammation.

3.2 Storytelling Design

We developed a first version of our immunology VR experience that focused on using interactivity to engage learners [8]. We found in an experiment that interactivity was helpful in improving engagement, but that no learning gains were associated with increased interactivity [9]. Thus, we designed a second version that emphasized rich storytelling along with interactive elements.

3.2.1 Story structure

Although storytelling is a frequently used design element in many educational applications, stories created from it differ a lot in their structures. At a very basic level, a storytelling experience can be just a voiceover narration that guides the learners through the experience with necessary learning information while they are

using the application. In contrast, a more engaging storytelling experience may present the learners with a dramatic story that contains some essential elements like characters, dialogue, and a story arc. We believe a well-structured dramatic story with characters engages learners more into the story world and thus helps them remember and retain more information from the learning experience.

To create a short (15 -20 minutes), dramatic story with embedded immunology concepts, we looked into traditional storytelling models for films and plays and adopted them into a story structure that met our instructional needs. We used screenplay-style writing as a main format to compose the story script due to its flexibility with rich character development and abundant dialogue. The classic Aristotle's three-act story arc and Syd Field's extended Aristotle model [10] for screenplay are the main models we referenced when developing our own story structure.

3.2.2 Characters

Characters are also important storytelling elements in the development of a story arc. Multi-character storytelling experience may have some instructional advantages: first, dialogue between characters is more interesting than monologues and third-person voiceover narration, thus engaging the learners more into the story. Second, dialogue between characters can be custom scripted to convey more information, such as embedded learning concepts, without much dullness. Based on the learning concepts and story plot, we developed the following characters for our story (Figure 1).

Protagonist:

- PHIL: a young neutrophil, embodied by the user

Antagonist:

- THE MUTANT: a mutated bacterium that is hard to kill

Other characters:

- MONO6/MAC: a monocyte who later differentiates into a macrophage
- MAJOR 16: a memory T Cell that recruits other white blood cells to infection site
- ARA-C: a virtual assistant of PHIL
- S.AUREUS: a bacterium that causes infection

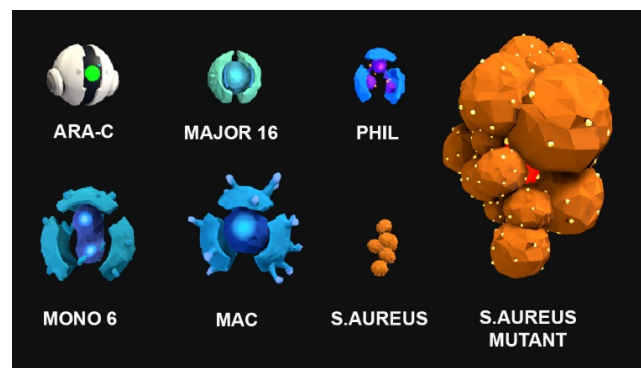


Figure 1: Characters from the story

3.2.3 Story Synopsis

A young neutrophil (the protagonist, embodied by the user) is dispatched to a site of infection caused by *S. Aureus* bacteria (non-player characters, controlled by game AI). While it learns how to use its three unique weapons (degranulation, phagocytosis, NETS) to fight the bacteria, it detects a dangerous mutant of *S. AUREUS* (the main antagonist, a non-player character) and decides to trace it

down and stop it from growing and spreading in the host body. Working with its virtual assistant (a non-player character, controlled by game AI) and a macrophage (a non-player character, controlled by game AI), the neutrophil finally traps the bacterium mutant with its deadly but suicidal weapon, NETS, by sacrificing itself. In doing so, it allows the macrophage to successfully process the bacterium mutant's antigen information and pass it on to many other immune cells, thus activating a full-body immune response against the bacteria mutants (Figure 2).



Figure 2: (Top) A macrophage character is engulfing a bacterium character. (Bottom) The user embodies the main character and fights the boss bacterium character.

3.3 Integration of Interactivity into a Dramatic Story

3.3.1 A guided interactive storytelling model

Our previous study on level of interactivity on VR learning [9] suggested that higher level of interactivity in an educational VR experience contributes to higher learning engagement and more experience enjoyment. While the storytelling is the main component of the experience to explain complex immunology concepts to the students, adding rich interactivity to the experience allows them to apply what they have learned to various tasks and reinforces their understanding of the concepts. We referenced and experimented with a guided interactive narrative approach proposed by Weiß and Müller [11] to prioritize the storyline, the plot, and the overall linear path of the story, but at the same time give the learners superposed interactivity at localized scenes (Figure 3). Specifically, the interactivity we integrated into the story combines system-controlled virtual travel and full user-controlled task performance and localized virtual movement.

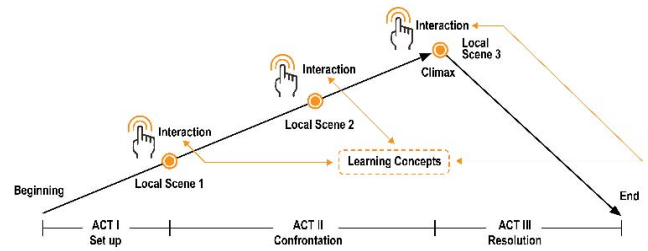


Figure 3: Integration of interactivity into a three-act story

3.3.2 Implementation

To develop the actual working prototype of the experience in VR, we took advantage of some new cinematic and animation tools available in the game engine Unity since its 2017.4 version and customized them to meet our design needs. Specifically, we used the Timeline tool in Unity to sequence all audio and dialogue in the experience. We then used Timeline Signals to trigger interaction events whenever we wanted, either during an audio clip that is playing or after it is played. Once an interaction event meets a certain condition, it can be scripted to trigger Timeline audio so that the Timeline resumes playing and the story progresses. Such a design setup allows us to create a system that responds precisely to the user's interactions and provides quick feedback to user activities, thus optimizing the VR learning experience. In addition, to create engaging character dialogues with emotions, we invited experienced student voice actors from Cinema and Performing Arts majors at VT to record the characters' dialogue.

3.4 3D Interaction and UI Design

3D interaction and UI design are a key component of our interactive storytelling experience. In general, we based our design strategies on previous studies of 3D interactions and user interface design [12] and used metaphors from real-world and popular culture to inspire the 3D UI designs in the application. The main advantage of using such a strategy is that the users are probably already familiar with real-world interactions and their interfaces and it may take them minimum time to learn how to use the UIs, which reduces their cognitive load and allows them to only focus on the learning experience. We describe some design examples in Table 1.

Table 1. 3DUIs and their real-world and pop culture metaphors

Interactions and 3D UIs	Metaphors	Design Rationale
Neutrophil embodiment	Airplane cockpit	Familiarity with dashboards and vehicle control
Neutrophil rolls on blood vessel surface	<i>Jurassic World</i> gyrosphere vehicle	Visual reference to reduce discomfort
Opening in the blood vessel	Magic portal in sci-fi movies	Transmigration is like using a portal
Neutrophil's three killing mechanisms	Weapon control and switch panel	Button selections in real-world devices

4 PROPOSED EXPERIMENT

With our immunology VR storytelling project, we hypothesize that both interactivity and storytelling may affect the students' learning outcomes from using the application. Using our current working prototype described in the paper, we propose a study to investigate how different levels of storytelling (no story, basic, advanced) affect students' VR learning outcomes. Specifically, we will create three experiment conditions in which the VR experiences mainly differ in their story structures and character setup. Our hypotheses are:

- Stories with higher levels of character setup and structures will engage the learners more in an educational VR experience.
- The learners will retain intended learning concepts from the advanced level of story for the longest period of time compared to lower levels of storytelling.

The study is currently in the phase of user studies and data collection. We will report our findings in a future paper.

5 CONCLUSION

Immersive storytelling in VR is a potential new instructional tool for science education. However, most current applications are not designed and optimized for learning and instructional purposes, being limited with only passive viewing experiences or lacking in well-developed story structures. In our design experiment, we explored a novel solution by integrating high interactivity and a well-structured story into an immersive learning experience. Our intent is to use interactivity and dramatic stories to engage the students into a deep level of information processing and retention for the embedded learning concepts. Our demo of a working prototype of the application is the first step of our on-going empirical research on how different levels of storytelling in an immersive VR storytelling experience affect science learning. We believe our work will contribute to the body of knowledge on how to design storytelling-based VR science learning experiences and optimize learning outcomes.

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